

Albian SALIHU

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Swiss national · French (native) · Albanian (native) · English (C1)

Education

EPFL, École Polytechnique Fédérale de Lausanne, Lausanne Sept **2017** – Sept **2024**
MSc Robotics (minor Space Technologies) · BSc Microengineering · Preparatory year (CMS)

CPLN, Centre Professionnel du Littoral Neuchâtelois, Neuchâtel Sept **2012** – Aug **2017**
Federal Vocational Baccalaureate · Federal VET Diploma (CFC) in Electronics

Professional experience and projects

Wasteflow — Robotics & Systems Integration Engineer May **2025** – Present

- Piloting an industrial PoC at a paper-sorting facility — sole technical owner from first integration meetings through on-site commissioning and iterative testing; target: 10% improvement in manual sorter performance.
- Coordinated PLC replacement at the facility to enable conveyor speed control; developed real-time algorithms adjusting a conveyor chain based on upstream contamination data from the existing vision infrastructure.
- Built a lab conveyor testbed replicating sorting conditions for repeatable controlled experiments before each field deployment.
- Trained and deployed waste-classification models for live decision-making; iterated using controlled lab runs and field data.
- Exploring robotic manipulation concepts for advanced sorting use cases; produced a demonstrator video to gauge industrial interest.

Swiss MotionTech — Master Thesis Feb **2024** – Sept **2024**

- Designed a real-time closed-loop feedback system correcting for multi-hardness silicone 3D printing. Validated on prosthetic liners in production.

EPFL Spacecraft Team — Semester Project Sept **2023** – Jan **2024**

- Designed and implemented a filament controller in VHDL on a Xilinx Zynq UltraScale+, regulating the thermionic emitter of a CubeSat TOF mass spectrometer for the ESA-supported CHESS mission, in collaboration with the University of Bern.

EPFL Robot Competition 2023 — Team member Feb **2023** – Jun **2023**

- Designed and built a fully autonomous robot (ROS2, SLAM, computer vision) achieving full task completion at the EPFL competition.

Technical and computer skills

- **Core strengths:** Real-time control systems, FPGA development (Xilinx Zynq SoC / Vivado / Vitis, Intel Quartus), embedded systems, computer vision and AI integration.
- **Robotics & Control:** ROS 2, UAV control theory, sensor integration, PLC integration, system-level integration and field testing.
- **AI & Data:** Machine learning and deep learning pipelines, model training and deployment for industrial vision tasks.
- **Electronics & CAD:** PCB design and prototyping (Altium Designer), circuit assembly (soldering, crimping), CAD tools (Fusion 360, Catia, Inventor, FreeCAD).
- **Programming:** Python, C/C++, VHDL, MATLAB, C#, Java.

Interests

Astronomy — I follow space missions and planetary science closely; I enjoy stargazing on clear nights.

Trekking — Regular hikes with my dog, good for the head after a week in front of screens.

DIY — Whether it's electronics, furniture, or whatever breaks first, I prefer to understand and fix things myself.